Lecture 5-Part 2

- Packet Loss and Throughput
- Network Layering

Application Layer Protocols

Lecture notes are based, in part, on slides by Kurose and Ross textbook “Computer Networking: A Top-Down Approach”, 8th edition

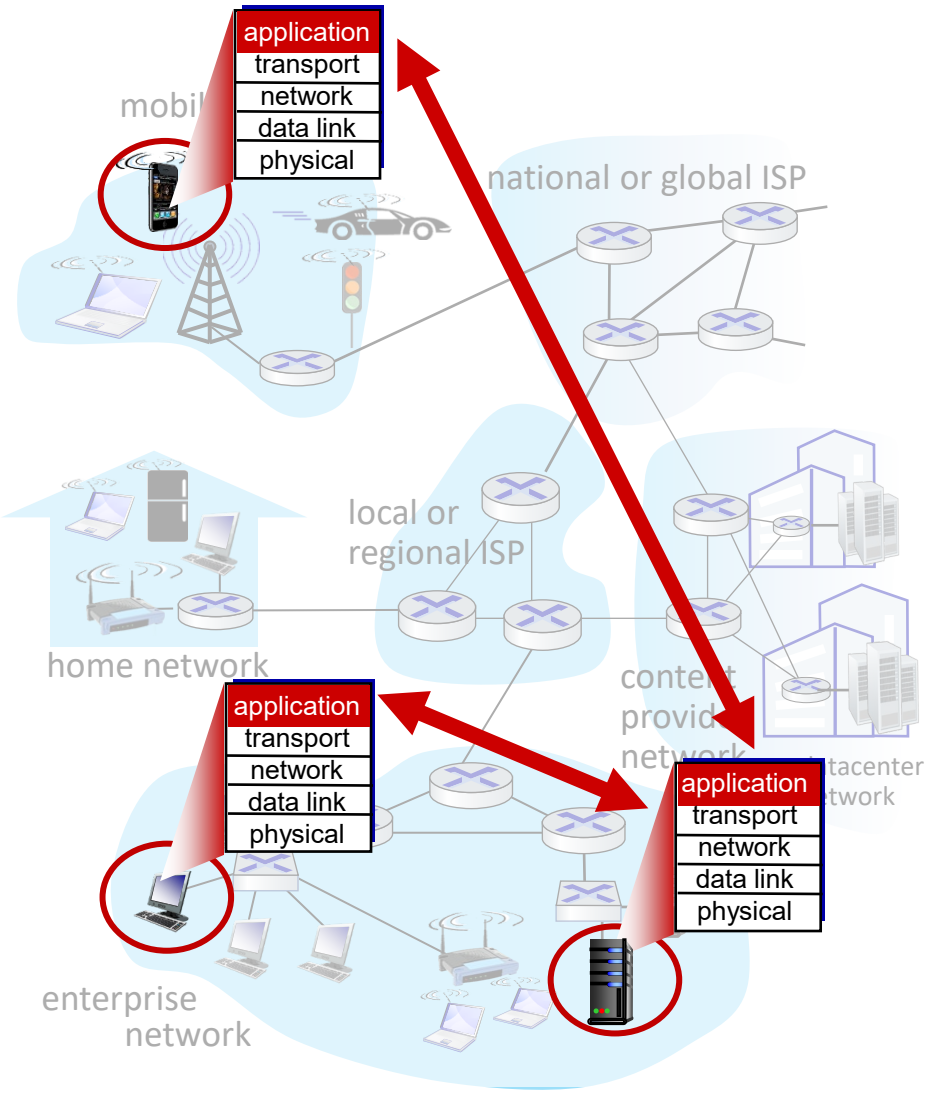
Application layer: Outline

- Overview
- Examples of Internet Applications
  - Web and HTTP
  - E-mail, SMTP, IMAP
  - The Domain Name System DNS
  - P2P applications
  - video streaming and content distribution networks
- Socket Programming

Overview: Network Applications

- Network Applications: Distributed processes
  - Running on network hosts in "user space"
  - communicate over network
  - e.g., email, file transfer, the Web

- Application layer protocol
  - Define messages exchanged by network apps and actions taken
  - Rely on services provided by lower layer protocols



Client and Server Process

Process: program running within a host

- Within the same host, two processes communicate using inter-process communication (defined by OS)
- Processes running in different hosts communicate with an application-layer protocol

Clients & servers Process

client process: process that initiates communication

server process: process that waits to be contacted

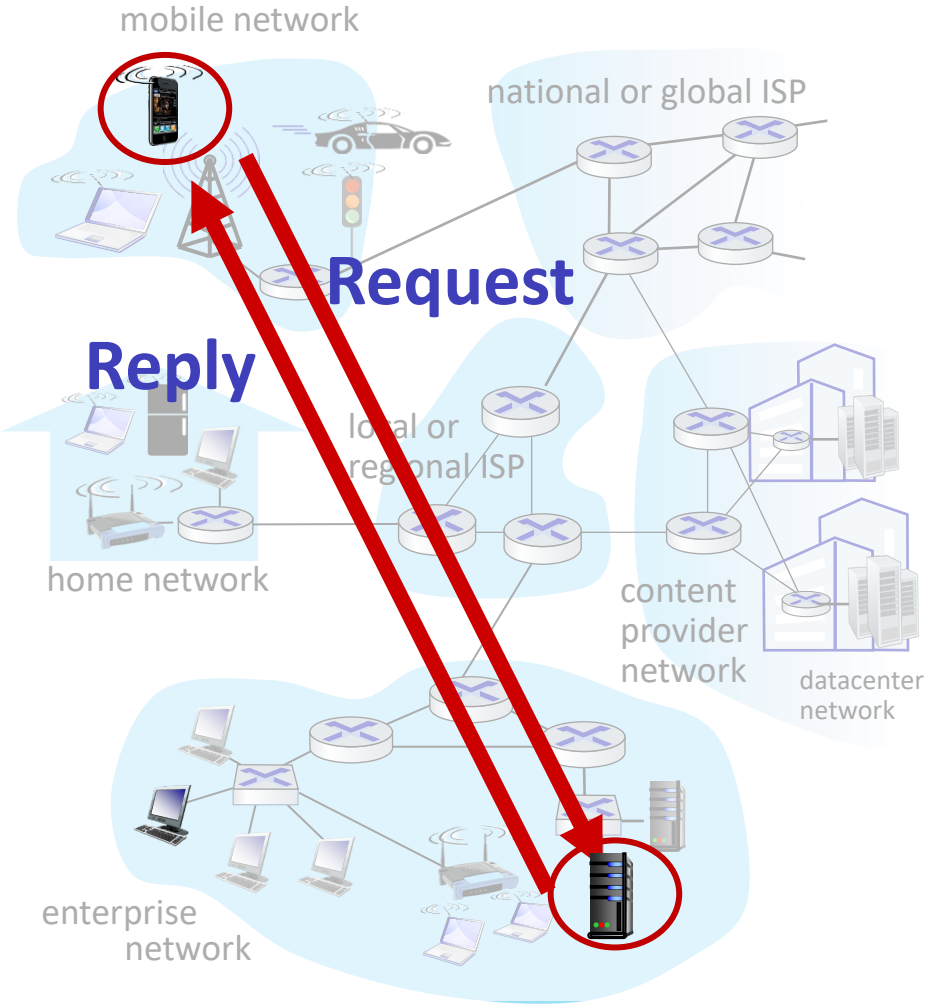
Client-server Architecture

Server:

- Always on
- permanent IP address
- provides requested service to client
- e.g., Web server sends requested Web page

Client:

- Initiates contact with server; typically requests service from server,
- for Web, client is implemented in browser



Peer-to-peer (P2P) architecture

- *no* always-on server
- end systems directly communicate
- peers request service from other peers, provide service in return to other peers
  - *self scalability* – new peers bring new service capacity, as well as new service demands
- peers are intermittently connected and change IP address

- Applications with P2P architectures have both client & server processes

